MINJOON PARK

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EDUCATION

University of California, San Diego **- June 2028 (expected)**

- GPA: 3.85/4.0
- B.S. in Chemical Engineering, B.S. in Molecular and Cell Biology (*Double Major*)
- Relevant Coursework: Reaction Engineering, Thermodynamics, Cell Biology, Fluid Mechanics, Organic Chemistry, Biochemical Structures, Genetics, Material Balances.

RESEARCH & INDUSTRY EXPERIENCE

University of California, San Diego **Fall 2025 - Current**

Undergraduate Researcher - NanoBioElectronic Lab (PI: Prof. Joseph Wang)

- Co-engineered noninvasive electrochemical biosensors (e.g. sweat and saliva cortisol), utilizing immobilized aptamer systems for continuous biomarker monitoring.
- Executed microfabrication protocols and evaluated sensor kinetics, temporal drift, and signal degradation across 20+ wearable human trials and 3 *in vivo* rodent models.

Korea Electronics Technology Institute (KETI) **Summer 2025**

Laboratory Intern - Advanced Batteries Research Center (PI: Dr. KyungSu Kim)

- Engineered a custom Java and ImageJ plugin for microscopy image processing, automating fast and efficient elemental distribution analysis across microstructural surfaces.
- Operated advanced characterization equipment (XRM, SEM-EDS) in cleanroom and dry-room environments for solid-state battery fabrication and electrolyte synthesis.

Columbia University **Spring 2023 - Spring 2024**

Remote Research Assistant - Yang Research Group (PI: Prof. Yuan Yang)

- Developed Python and MATLAB automation pipelines to optimize geometric variables, accurately modeling angular emissivity parameters for dynamic thermal regulation systems.
- Performed thermodynamic simulations in EnergyPlus across 16 U.S. climate zones, analyzing datasets to demonstrate ~24% annual HVAC energy savings compared to conventional baselines.

Dalhousie University **Fall 2022 - Spring 2024**

Remote Research Assistant - PHX Group (PI: Prof. Penghao Xiao)

- Conducted Density Functional Theory simulations on the Cedar Supercomputer cluster (30,000+ core-hours) to model atomic-scale kinetics in multi-phase materials.
- Simulated lattice defects to determine how structure disruptions drive charge-discharge kinetics.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, MATLAB, Java, R, JavaScript, MS Excel.
- **Laboratory & Engineering Techniques:** Microfabrication, Electrochemical Analysis, SEM-EDS, Biochemical Lab Methods (SDS-PAGE, Western Blot, Spectroscopy).
- **Computational & Modeling Tools:** OriginPro, VASP (DFT), EnergyPlus, OpenStudio, SketchUp, BioRender, SolidWorks, ImageJ.

CO-AUTHORED PUBLICATIONS

1. Sabbagh, Barak, et al. *"Wearable Cortisol Detection in Sweat." ACS Sensors, 2026.*
2. Cheng, Qilong, et al. *"Tunable Angular Emissivity for Thermal Regulation." Cell Rep. Phys. Sci., 2024.*
3. Xiao, Penghao, et al. *"Charge-Discharge Kinetics in Off-Stoichiometric $LiNiO_2$." Chem. Mater., 2024.*

ACTIVITIES

- Cellist - UCSD Chamber Orchestra
- Secretary - UCSD LEGO Club
- Other Interests: Acrylic Painting, Endurance Running, Lap Swim